

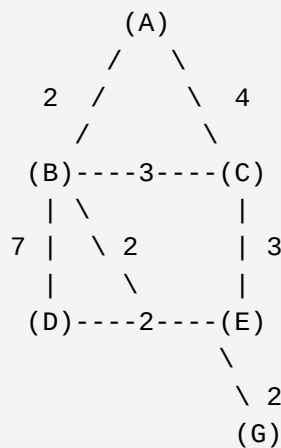
Artificial Intelligence Laboratory Assignment

Problem Statement — Search Algorithms

Student	Darwin Jacob
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Topics Covered	Informed Search (A*), Adversarial Search (Minimax, Alpha-Beta Pruning)
Total Marks	20 (Q1: 10 Marks, Q2: 10 Marks)

Question 1 : Dry Run of A Search Algorithm (10 Marks)

Consider the following weighted, undirected graph:



Heuristic Values $h(n)$ (estimated straight-line distance to Goal G)

Node	A	B	C	D	E	G
$h(n)$	7	6	4	3	2	0

Task

Perform a dry run of the A* Search algorithm to find a path from node **A** (start) to node **G** (goal). For each iteration, show:

- Current node being expanded
- Contents of the Open List
- Contents of the Closed List
- $g(n)$ — actual cost from start to n
- $h(n)$ — heuristic estimate from n to goal
- $f(n) = g(n) + h(n)$ — total estimated cost

Finally, the solution must:

- 1 Identify the optimal path from A to G.
- 2 Calculate the total path cost of the optimal path.

Question 2 : Dry Run of Minimax with Alpha-Beta Pruning (10 Marks)

Consider the following two-ply game tree (evaluated left to right):



Task

Perform a dry run of the Minimax algorithm with Alpha-Beta Pruning, evaluating the tree strictly from left to right. For each node, show:

- Alpha (α) value at each node
- Beta (β) value at each node
- Selected / backed-up value at each node
- Any node(s) that get pruned, with justification

Finally, the solution must:

- 1 Determine the best move for MAX.
- 2 State the final minimax value of the game tree.
- 3 List all pruned nodes (if any).

Deliverables for this assignment: Problem_Statement.pdf, Solution.pdf, search_algorithms.py, output.png, Report.pdf, README.md